Editor - C:\Users\CISCO PC-13\Desktop\labo\hola.m *

untitled | hola.m * | chau.m | +

```matlab
clc;
clear;
% Funcion
f = @(x) exp(x);
% Punto de evaluacion
x = 2;
% Valores de h
h_values = [0.5 0.1 0.05 0.01 0.005 0.001];
% Valor exacto
exacto = exp(2);
fprintf('VALOR EXACTO = %.15f\n\n', exacto);
%  TABLA CON 6 DIGITOS
fprintf('PRECISION 6 DIGITOS\n');
fprintf('\n');
fprintf(' h\t\tf''(2)\t\t\tError\t\t\tf''''(2)\t\tError\n');
for h = h_values
    % Aproximacion de primera derivada
```

Command Window

```
VALOR EXACTO = 7.389056098930650

PRECISION 6 DIGITOS

 h       f'(2)            Error            f''(2)       Error
 0.500       7.700805        3.117488e-01     7.544283         1.552272e-01
 0.100       7.401377        1.232125e-02     7.395216         6.159600e-03
 0.050       7.392135        3.079158e-03     7.390596         1.539515e-03
 0.010       7.389179        1.231516e-04     7.389118         6.157566e-05
 0.005       7.389087        3.078777e-05     7.389071         1.539387e-05
 0.001       7.389057        1.231509e-06     7.389057         6.160042e-07


PRECISION 8 DIGITOS

 h       f'(2)            Error            f''(2)       Error
 0.500       7.70080489   3.11748791e-01   7.54428333   1.55227234e-01
 0.100       7.40137735   1.23212525e-02   7.39521570   6.15959963e-03
 0.050       7.39213526   3.07915824e-03   7.39059561   1.53951497e-03
 0.010       7.38917925   1.23151551e-04   7.38911767   6.15756593e-05
 0.005       7.38908689   3.07877721e-05   7.38907149   1.53938720e-05
 0.001       7.38905733   1.23150872e-06   7.38905671   6.16004188e-07
>>
```

Editor - C:\Users\CISCO PC-13\Desktop\labo\chau.m *

chau.m * | +

```matlab
% Datos
x = [1.5 1.9 2.1 2.4 2.6 3.1];
y = [1.0628 1.3961 1.5432 1.7349 1.8423 2.0397];
% Punto donde evaluar
x0 = 2.25;
% Polinomio interpolante
p = polyfit(x,y,length(x)-1);
disp('Polinomio interpolante:')
disp(p)
dp = polyder(p);
fp = polyval(dp,x0);
ddp = polyder(dp);
fpp = polyval(ddp,x0);
% Resultados
fprintf('\nRESULTADOS:\n');
fprintf('f''(2.25)   = %.10f\n',fp);
fprintf('f''''(2.25) = %.10f\n',fpp);
```

Command Window

```
Polinomio interpolante:
   -0.0014     0.0311    -0.2233     0.5236     0.4387    -0.1666


RESULTADOS:
f'(2.25)   = 0.6393496232
f''(2.25)  = -0.4013578681
>>
```